



Importance of Lysosomal Trapping and *Plasmodium* Parasite Infection on the Pharmacokinetics of Pyronaridine: A Physiologically Based Pharmacokinetic Model-Based Study

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Abstract

Background and Objective Pyronaridine is a blood schizonticide with a high blood-to-plasma ratio, effective against *Plasmodium* parasites. As a lipophilic, moderately strong base, it accumulates in low-pH compartments such as lysosomes and parasite food vacuoles, leading to tissue accumulation and differences in drug exposure between healthy individuals and patients with malaria. This study applied physiologically based pharmacokinetic (PBPK) modeling to evaluate the effects of lysosomal sequestration, red blood cell (RBC) accumulation, and parasitemia on pyronaridine pharmacokinetics.

Methods Data were available from a phase I clinical trial and the PYRAPREG study. PBPK models were developed in PK-Sim® and MoBi®. A standard multicompartment structure was expanded by adding lysosome compartments to relevant organs. To account for malaria infection, *Plasmodium* parasite compartments were incorporated into RBCs, with volume scaled by parasitemia.

Results Data from 52 healthy individuals and 25 patients with malaria were used for model optimization. Incorporating lysosomal sequestration was essential for capturing pyronaridine distribution. In patients with malaria, incorporating low hemoglobin (Hb) and drug accumulation in the parasite compartment enabled an adequate description of whole blood pharmacokinetics. Simulations showed that free pyronaridine concentrations in the parasite compartment were over 10-fold higher than that in whole blood. Higher parasitemia was associated with increased area under the curve (AUC)_{0–24h} and C_{max}, mainly on day 1, as parasitemia decreased rapidly. However, the subsequent decrease in Hb had the opposite effect, lowering AUC_{0–24h} and C_{max} on the following days.

Conclusions This study demonstrates the value of PBPK modeling in elucidating key pharmacokinetic mechanisms, revealing the critical roles of lysosomal sequestration, Hb level, and parasitemia in pyronaridine disposition.

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Key Points

This study used PBPK modeling to investigate how lysosomal sequestration and malaria-associated pathophysiological factors, such as low hemoglobin levels and elevated parasitemia, affect the pharmacokinetics of pyronaridine.

The model identified lysosomal sequestration as a key driver of extensive tissue distribution, while drug accumulation in parasite compartments and changes in hemoglobin levels substantially impacted both systemic and target site exposure.

These findings demonstrate the value of PBPK modeling in elucidating the complex distribution mechanisms of pyronaridine.

1 Introduction

Malaria, caused by *Plasmodium* parasites infecting red blood cells (RBCs), continues to pose a global health challenge, especially in tropical and subtropical regions. Despite ongoing efforts to control and eliminate the disease, the rise of drug-resistant strains has complicated treatment strategies [1]. Pyronaridine-artesunate, the most recent addition to the World Health Organization (WHO)-recommended artemisinin-based combination therapies (ACTs) for uncomplicated *P. falciparum* and *P. vivax* malaria, offers a promising alternative in areas where resistance to other antimalarials is a growing concern [2].

Pyronaridine serves as the long-acting component in this ACT, playing a crucial role in clearing residual parasitemia after the artemisinin component has been eliminated. Although it has been used as an antimalarial for decades, its clinical pharmacology and mechanism of action are not fully understood [3]. As a lipophilic, moderately strong base with four ionizable amines ($\text{pK}_a > 6$) [4], pyronaridine preferentially accumulates in low-pH cellular compartments such as lysosomes [5, 6]. This lysosomal sequestration may result in extensive drug accumulation in, and slow release from, lysosome-rich tissues such as the lungs, liver, kidneys, and spleen [7]. These properties may explain findings from prior population pharmacokinetic (PK) studies, which reported a peripheral volume of distribution exceeding 6000 liters in both healthy individuals and patients with malaria, including adults and children [8, 9]. Moreover, lower drug exposure has been observed in patients with malaria

compared with healthy individuals receiving comparable dosing regimens, suggesting that malaria infection may influence the disposition of the drug [3].

Pyronaridine is a highly effective blood schizonticide, exhibiting potent activity against the asexual blood stages of the *Plasmodium* parasite. Its mechanism of action involves inhibiting hemozoin formation within the parasite's digestive vacuole, thereby preventing the detoxification of heme, a process similar to that of other quinolone-based antimalarials [10]. The pharmacological characteristics of the drug enable efficient delivery to the target site, with a high blood-to-plasma ratio supporting its distribution within the bloodstream [4]. The acidic environment of the parasite's digestive vacuole (pH 5.1) promotes selective pyronaridine accumulation at its site of action [10]. In patients with malaria, factors such as low hemoglobin (Hb) and high parasitemia may alter drug distribution and elimination [11], complicating its PK and therapeutic profile.

This study aimed to evaluate the role of lysosomal sequestration in the distribution of pyronaridine to peripheral tissues and to explore how low Hb and variable parasitemia influence its distribution to RBCs in patients with malaria. To address these objectives, we employed a physiologically based pharmacokinetic (PBPK) modeling approach, incorporating a mechanistic model to account for lysosomal sequestration and the pathophysiological effects of malaria infection.

2 Method

2.1 Pharmacokinetic Data

2.1.1 Healthy Individuals

Pyronaridine PK data from healthy individuals were kindly provided by Shin Poong Pharm. (Seoul, Korea). Data were available from the single ascending dose (SAD) and the multiple ascending dose (MAD) studies of a phase I clinical trial (SP-C-001-03), conducted at the Clinical Trial Center of Seoul National University, Seoul, South Korea [12]. Participants received pyronaridine tetraphosphate-artesunate at doses of 6+2 mg/kg, 9+3 mg/kg, 12+4 mg/kg, or 15+5 mg/kg, either as a single dose or once daily for three consecutive days. The administered pyronaridine tetraphosphate dose corresponds to 56% pyronaridine base. Accordingly, a 60 kg individual in the respective dose groups received 200, 300, 400, or 500 mg of pyronaridine free base. Details of the study design have been published elsewhere [12].

Whole blood samples were collected at 0.33, 0.68, 1, 1.33, 1.67, 2, 2.5, 3, 4, 5, 8, 12, and 24 h, as well as on days 2, 3, 5, 7, and 10 after the first dose in the SAD study and after the third dose in the MAD study. Additional samples were collected prior to the second and last dose administration in the MAD study. Pyronaridine concentrations in whole blood were measured using a validated liquid chromatography–tandem mass spectrometry (LC-MS/MS) assay with a quantification range of 5.7–855 µg/L [13].

2.1.2 Patients with Malaria

Data from patients with malaria were available from the PYRAPREG study (PACTR202011812241529), a phase III, noninferiority, randomized, open-label clinical trial involving pregnant women with malaria in five malaria-endemic countries in sub-Saharan Africa [14]. The PK data included in this analysis are from a cohort of nonpregnant, human immunodeficiency virus (HIV)-negative women, recruited in the Democratic Republic of the Congo and in Mozambique. Patients with *P. falciparum* mono-infection confirmed by microscopy at any density and Hb concentration ≥ 7 g/dL were included. Patients received Pyramax® tablets (180 mg pyronaridine tetraphosphate + 60 mg artesunate per tablet) once daily for 3 days. The dose was based on body weight: two tablets for 24 to < 45 kg, three tablets for 45 to < 65 kg, and four tablets for ≥ 65 kg, providing 200–400 mg pyronaridine free base.

Whole blood samples were collected at 1, 2, 6, 10, and 24 h after the first dose. Additional samples were taken prior to the last dose and on study days 7, 14, 21, 28, and 42. Pyronaridine concentrations in whole blood were measured using a validated ultra-performance LC-MS/MS assay with a quantification range of 0.5–500 µg/L [15].

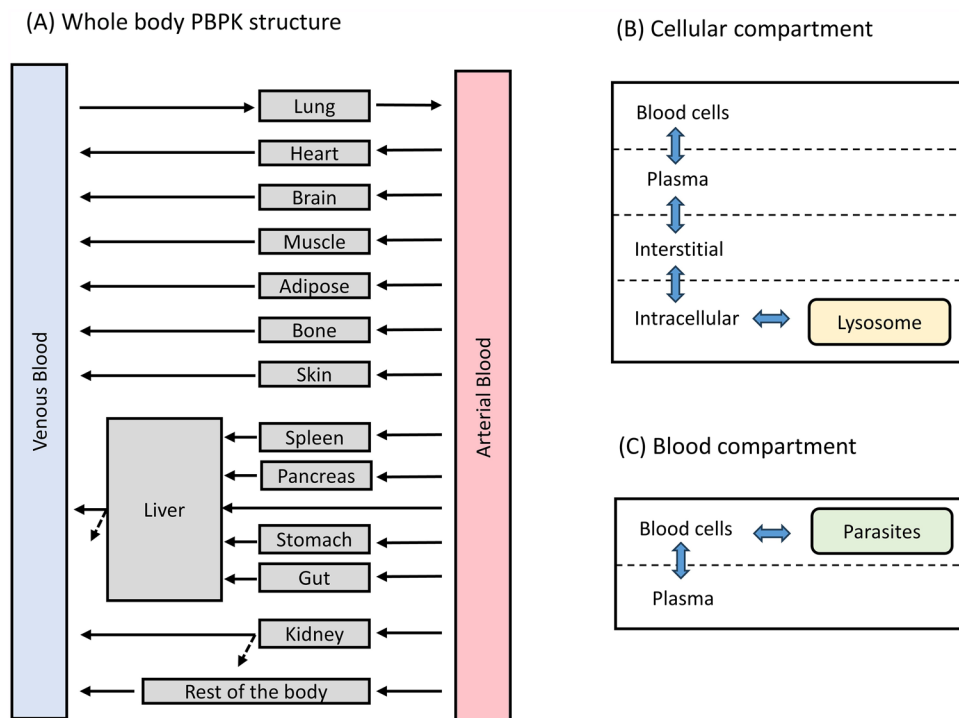
2.2 PBPK Model Development

2.2.1 General Workflow

Whole-body PBPK models for pyronaridine were developed in PK-Sim® and MoBi® (Open Systems Pharmacology software, version 11.3) [16], to describe the PK of pyronaridine in typical healthy individuals and patients with malaria on the basis of anatomical and physiological parameters. The model consists of interconnected compartments representing major organs and tissues (e.g., liver, kidney, heart, lung, stomach, Intestine, spleen, muscle, fat, skin, and brain), each subdivided into vascular (blood cells and plasma), interstitial, and intracellular spaces to capture drug distribution at the tissue level (Fig. 1A).

Tissues are linked via arterial and venous blood compartments, with drug transport governed by blood flow, passive diffusion, or transporter-mediated processes. Distribution is classified as perfusion- or permeability-limited on the basis of physicochemical properties. Metabolism occurs primarily in the liver, while renal clearance accounts for elimination. System-specific

Fig. 1 Physiologically based pharmacokinetic (PBPK) model structure



parameters, including organ volumes, blood flows, tissue partitioning, and enzyme expression levels, are integrated to mechanistically predict drug PK.

Individual-specific input parameters, including body weight, height, body mass index (BMI), and age, were based on population medians used for model evaluation and optimization. Protein expression profiles were based on whole-genome expression array data from ArrayExpress [17]. Compound-specific parameters, encompassing physicochemical properties related to absorption, distribution, metabolism, and excretion (ADME) processes of pyronaridine were derived from literatures. Owing to the lipophilic and basic nature of pyronaridine, tissue-to-plasma partition coefficients ($K_{p,organ}$) were calculated using Rodgers and Rowland model for moderate-to-strong bases [18, 19], and the cellular permeability was calculated using the charge-dependent Schmitt model [20], both under the framework implemented in PK-Sim® and MoBi®.

In the Rodgers and Rowland model, $K_{p,organ}$ was described as the following equation:

$$K_{p,organ} = \left(\frac{1 + X \times f_{IW}}{1 + Y} \right) + f_{EW} + \left(\frac{K_a \times [AP]_T \times X}{1 + Y} \right) + \left(\frac{P \times f_{NL} + (0.3P + 0.9) \times f_{NP}}{1 + Y} \right), \quad (1)$$

where P is the n -octanol:water partition coefficient for the unionized compound in tissues, f represents the fractional tissue volume, with subscripts IW and EW referring to intra- and extracellular tissue water, respectively, and NL and NP denoting tissue neutral lipids and neutral phospholipids. $[AP]_T$ represents the tissue concentration of acidic phospholipids, and K_a is the drug's affinity constant for acidic phospholipids.

For polyprotic base, X and Y in Eq. 1, accounting for drug ionization, were defined as follows:

$$X = 10^{pKa2 - pH_{IW}} + 10^{pKa1 + pKa2 - 2pH_{IW}} \quad (2)$$

$$Y = 10^{pKa2 - pH_p} + 10^{pKa1 + pKa2 - 2pH_p}, \quad (3)$$

where $pKa_1 < pKa_2$, and pH_p and pH_{IW} represent the pH of plasma (≈ 7.4) and organ-specific intracellular tissue water, respectively.

Following the Rodgers and Rowland model, drug partitioning between plasma and RBCs ($K_{p,RBC}$) was calculated as follow:

$$K_{p,RBC} = \frac{HCT - 1 + BP_{ratio}}{HCT}, \quad (4)$$

where hematocrit (HCT) and blood-to-plasma concentration ratio (BP_{ratio}) were used.

Model development followed a three-step process. First, a PBPK model for healthy individuals receiving 6, 9, 12, and 15 mg/kg pyronaridine tetraphosphate was built on the basis of literature values. A standard healthy individual profile implemented in PK-Sim® (ICRP 2002) [21] was used, assuming a body weight of 60 kg and a Hb level of 13.6 g/dL. Second, lysosome compartments were added to the structural model of healthy individuals to assess the impact of lysosomal sequestration. Finally, the model developed for healthy individuals was applied to patients with malaria, receiving 540 or 720 mg pyronaridine tetraphosphate, with parasite compartments incorporated into the blood organs. To represent a typical patient with malaria, the model was adjusted for a reduced Hb level of 8 g/dL and a parasitemia of 2%, reflecting characteristics of uncomplicated malaria [2, 22]. In all models, the administered drug dose was implemented as a pyronaridine base.

2.2.2 Lysosome Compartment

To the model structure as described, an additional lysosome compartment was incorporated into relevant organs (Fig. 1B), following the method provided by ESQlabs GmbH [23]. These organs included the brain, fat, gonads, heart, kidneys, large, and small intestines (including their mucosal compartments), lungs, muscles, pancreas, skin, spleen, and stomach, as well as the pericentral and periportal zones of the liver. In each organ, the volume of lysosome compartment was set at approximately 2% of the intracellular volume, consistent with literature estimates indicating that lysosomes typically occupy 1–5% of the total intracellular volume in mammalian cells [24, 25]. This value was used as a representative average across tissues for modeling purposes. The lysosomal pH, which typically ranges from 4.5 to 5, was fixed at 4.51 (pH_{lys}) [5].

Between the intracellular space and lysosomes, the pronounced pH gradient was assumed to be the primary driving force for the disposition of pyronaridine. For model simplification, the interactions with neutral lipids, neutral phospholipids, and acidic phospholipids were omitted. Given that lysosomes lack an extra-organelle compartment, the intracellular tissue water fraction was assumed 1 [26]. Consequently, the partition coefficient for lysosomes ($K_{p,lys}$) was simplified from Eq. 1 as follows:

$$K_{p,lys} = \frac{1 + 10^{pKa2 - pH_{lys}} + 10^{pKa1 + pKa2 - 2pH_{lys}}}{1 + 10^{pKa2 - pH_{IW}} + 10^{pKa1 + pKa2 - 2pH_{IW}}}. \quad (5)$$

The mass transfer between the intracellular space and lysosomes was described by the following equation:

$$\frac{dA}{dt} = Q_{\text{cell,lys}} \times \frac{1}{K_{\text{p,organ}}} \times C_{\text{cell}} - Q_{\text{cell,lys}} \times \frac{1}{K_{\text{p,lys}}} \times C_{\text{lys}}, \quad (6)$$

where $Q_{\text{cell,lys}}$ represents the flow rate between intracellular space and lysosomes, and C_{cell} and C_{lys} denote the drug concentrations in the intracellular space and lysosomes, respectively.

2.2.3 Plasmodium Parasite Compartment

To account for patients with malaria, a *Plasmodium* parasite compartment was incorporated into organs with larger RBC volume (> 0.01 L), including arterial blood, bone, fat, kidney, liver periportal, lung, muscle, skin, spleen, and venous blood (Fig. 1C). The volume of the parasite compartment (V_{parasite}) was determined on the basis of HCT and parasitemia as follows:

$$V_{\text{parasite}} = V_{\text{blood}} \times \text{HCT} \times \text{parasitemia}. \quad (7)$$

Here, V_{blood} represents the volume (L) of the blood compartment in respective organ, with the volume of RBCs determined by the HCT (L/L). Parasitemia can be quantified as the proportion of malaria-infected RBCs relative to total RBCs, typically assessed by counting parasitized RBCs among 500–2000 RBCs in a thin blood smear [27]. Consequently, V_{parasite} serves as an approximation of the volume of infected RBCs, assuming a uniform RBC density of 1 g/mL.

In ACT, the potent artemisinin derivative rapidly decreases parasite density. For artesunate, the parasite clearance time to achieve a 99% reduction in parasitemia (PCT_{99}) has been estimated to be around 30 h [28, 29]. Since this rapid decline occurs early in treatment and was not captured, we accounted for the treatment effect by incorporating the expected dynamics of parasitemia as follows:

$$\frac{d\text{Parasitemia}}{dt} = -k_{\text{kill}} \times \text{parasitemia}, \quad (8)$$

where k_{kill} represents the parasite clearance rate, calculated at 0.15 h^{-1} using the following equation:

$$k_{\text{kill}} = -\frac{\ln(0.01)}{\text{PCT}_{99}}. \quad (9)$$

Owing to the acidic environment of the *Plasmodium* food vacuole, the pH of the parasite compartment ($\text{pH}_{\text{parasite}}$) was set to 5.1 [30]. The partition coefficient for parasite compartment ($K_{\text{p,parasite}}$) was derived using the same approach as Eq. 5, accounting for pH differences between the parasites and RBCs. Drug distribution between the parasites and RBCs was modeled similarly to Eq. 6, with $Q_{\text{RBC,parasite}}$

representing the flow rate between RBCs and parasites, and C_{RBC} and C_{parasite} denoting drug concentrations in the RBCs and parasites, respectively.

2.3 Model Evaluation and Optimization

The model was evaluated and optimized using PK observations from both healthy individuals and nonpregnant, HIV-negative women with malaria. Parameter identification was performed for flow rates ($Q_{\text{cell,lys}}$ and $Q_{\text{RBC,parasite}}$), owing to the absence of established values in the literature. In addition, for a subset of highly variable parameters, including lipophilicity, specific intestinal permeability, BP_{ratio} , and partition coefficients ($K_{\text{p,lys}}$ and $K_{\text{p,parasite}}$), parameter identification was performed. Parameter identification was carried out using Levenberg–Marquardt or Monte–Carlo algorithm with 1000 iterations to identify a global optimum.

Model performance was evaluated by comparing predicted and observed concentrations, with predictions deemed acceptable if they fell within a 2-fold prediction interval margin [31]. Additional data from the literature on comparable populations were used for external validation. For healthy individuals, validation used data from Thai subjects ($n = 15$) [32] and from a phase I relative bioavailability study comparing tablet and granule formulations ($n = 42$) [33], each receiving a single 540 mg dose of pyronaridine. For patients with malaria, data from the 9 mg/kg dose group in a phase II dose-finding study ($n = 5$) were used [34], corresponding to a 540 mg dose. All external data were digitized using WebPlotDigitizer [35].

2.4 Target Site Exposure and the Effect of Malaria Severity

Concentration-time profiles were simulated for various matrices, including whole blood, RBCs, plasma, and the parasite compartment, to assess the best proxy for target site exposure. These concentrations were compared with the *in vitro* half maximal inhibitory concentration (IC_{50}) values (mean $\pm$ SD: $10.95 \pm 8.30 \text{ } \mu\text{g/L}$) derived from a susceptibility study on *P. falciparum* isolates across multiple African countries [36], to evaluate PK target attainment.

To investigate the impact of malaria infection on the whole blood PK profile of pyronaridine, simulations were conducted incorporating variations in Hb and parasitemia. Simulations were performed for patients with Hb levels of 13 g/dL (no anemia) and 7 g/dL (severe anemia) [37], in combination with parasitemia levels of 1% (low parasite burden) and 5% (high parasite burden) [27].

Secondary PK parameters, including area under the concentration-time curve from time zero to infinity ($AUC_{0-\infty}$), AUC 24 h post-dose (AUC_{0-24h}), peak concentration (C_{max}), time to C_{max} (T_{max}), and half-life ($T_{1/2}$), were derived using noncompartmental analysis (NCA) via PKanalix® [38], on the basis of simulated concentrations data up to day 10.

3 Results

3.1 Pharmacokinetic Data

Data from 52 healthy individuals (28 from the SAD study and 24 from the MAD study) and 25 female patients with malaria were used for model evaluation and parameter identification. The demographic and baseline characteristics of the study participants are presented in Table 1. An overview of the physicochemical properties and ADME characteristics of pyronaridine, used as initial model inputs and estimated in the final model, is presented in Table 2. Blood-to-plasma ratios reported in the literature ranged from 2 to 15; an initial value of 3 was selected for model development [3]. As only Hb concentrations were available in the clinical datasets, HCT was approximated by

assuming it is typically three times the Hb value [39]. For the calculation of partition coefficients, the two strongest basic pKa values of 10 and 9.88 were used [40]. The model incorporated hepatic metabolism of pyronaridine via CYP3A4, CYP1A2, and CYP2D6 as the primary elimination pathways [41] (see Table 2).

3.2 Effect of Lysosome Ion Trapping and Malaria Infection

The initial simulation for healthy individuals, without incorporating lysosome compartments, failed to capture the extensive distribution of pyronaridine, overestimating the initial rapid decline and underestimating the prolonged slow elimination. Adding lysosome compartments substantially improved the model fit (Fig. 2), with $K_{p,lys}$ estimated at 9×10^5 , consistent with the calculations from Eq. 5. The flow rate $Q_{cell,lys}$ was fixed at a large value, assuming an instantaneous equilibrium between the intracellular space and lysosomes. The final model provided an adequate description of the whole blood pyronaridine concentration in healthy individuals who received 6, 9, 12 and 15 mg/kg pyronaridine tetraphosphate (Fig. 3). The average fold errors (AFE) for concentrations at these dose levels were 1.21, 1.32,

Table 1 Demographics (median, range)

	Healthy volunteers ^a (<i>n</i> = 52)	Patients with malaria (<i>n</i> = 25)
Country (<i>n</i>)		
South Korea	52	-
Democratic Republic of the Congo	-	23
Mozambique	-	2
Pyronaridine dose (<i>n</i>) ^b		
6 mg/kg	13	-
9 mg/kg	13	-
12 mg/kg	13	-
15 mg/kg	13	-
360 mg ^c	-	1
540 mg ^c	-	14
720 mg ^c	-	10
Body weight (kg)	61.5 (50–70)	61.0 (41–82)
Height (cm)	171.0 (154.0–183.0)	162 (152–175)
Body mass index (kg/m ²)	21.1 (18.1–23.8)	23 (17.3–30.5)
Female (%)	40.6%	100%
Hemoglobin (g/dL)	13.6 ^d	11.7 (8.6–13.6)

^aNumber derived from the summary table in the Phase I clinical trial report (SP-C-001-03)

^bDose represents the amount of pyronaridine tetraphosphate containing 56% of the free base

^cEquivalent to pyronaridine dose of 8.7–11 mg/kg

^dHemoglobin values are absent in the phase I clinical trial report. Therefore, a value of 13.6 g/dL, as provided by the PK-Sim® standard database, was used in the model

Table 2 Input parameters and final parameters for the whole-body physiologically based pharmacokinetic (PBPK) model of pyronaridine

	Input parameters	Final estimates	References
Molecular weight	518 g/mol	518 g/mol	[4, 40]
Solubility	357 mg/L (pH7.4)	357 mg/L (pH7.4)	
Lipophilicity (Log P)	5.6	5.41	
pKa	7.39, 9.88, 10.3	7.39, 9.88, 10.3	
Plasma protein binding	95%	95%	
Blood-to-plasma ratios	3	2.31	
Intestinal permeability	4.7×10^{-3} cm/s	7×10^{-2} cm/s	
Dissolution Weibull function ^d			
50% dissolved time	10 mins	10 mins	48–
Dissolution shape	1	1	-
Hemoglobin			
Healthy individuals	13.6 g/dL	13.6 g/dL	[37]
Patients with malaria	8 g/dL	8 g/dL	
Specific clearance (CL_{spec}) ^a			
CL_{spec} CYP1A2	0.66 L/ μ mol/min	0.79 L/ μ mol/min	[41]
CL_{spec} CYP2D6	2.97 L/ μ mol/min	1.04 L/ μ mol/min	
CL_{spec} CYP3A4	0.28 L/ μ mol/min	0.48 L/ μ mol/min	
Lysosome compartment			
Partition coefficient ($K_{\text{p,lys}}$) ^b	9.5×10^4	9×10^5	-
Flow rate ($Q_{\text{cell,lys}}$)	–	5 L/min (fixed)	-
Parasite compartment			
Partition coefficient ($K_{\text{p,parasite}}$) ^c	1.7×10^4	13	-
Flow rate ($Q_{\text{RBC,parasite}}$)	–	5 L/min (fixed)	-

^a*In vitro* CL for liver microsomes normalized to enzyme concentration^bpHs of intracellular space and lysosomes are defined as 7 and 4.51, respectively^cpHs of red blood cells and parasites are defined as 7.22 and 5.1, respectively^dData are not available from the literature; the value was approximated to represent the rapid oral absorption reported in previous studies**Table 3** Secondary pharmacokinetic (PK) parameters in patients with varying degrees of malaria-induced hemoglobin reduction and associated parasitemia

	Hb 7 g/dL Parasitemia 1%	Hb 7 g/dL Parasitemia 5%	Hb 13 g/dL Parasitemia 1%
$AUC_{0-24h,D1}$ (μ g·day/L)	98	150	110
$AUC_{0-24h,D3}$ (μ g·day/L)	142	143	153
AUC_{inf} (μ g·day/L)	797	851	801
C_{max} (μ g/L)	218	380	282
T_{max} (h)	50.3	2.25	50.3
$T_{1/2}$ (d)	4.38	4.38	4.54

Hb hemoglobin (g/dL), $AUC_{0-\text{inf}}$ area under the concentration-time curve from time zero to infinity, AUC_{0-24h} area under the concentration-time curve 24 h post-dose, C_{max} peak concentration, T_{max} time to peak concentration, $T_{1/2}$ half-life

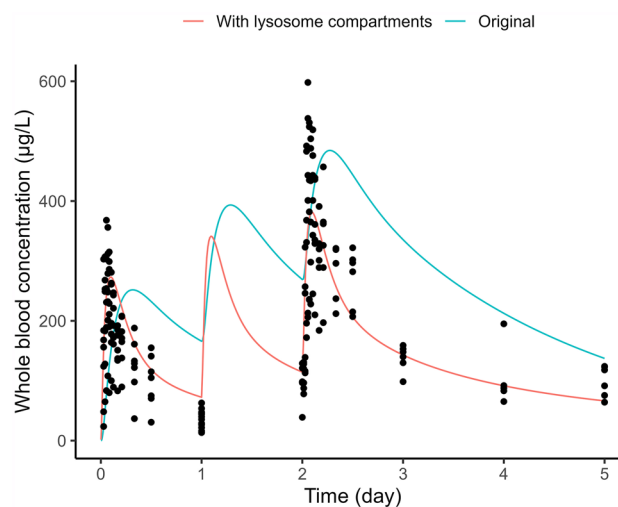
**Fig. 2** Effect of lysosomal sequestration. Model predictions with (red) and without (blue) the effect of lysosomal sequestration. The predictions are based on a typical healthy individual receiving 540 mg of pyronaridine tetraphosphate. Black dots represent observations from phase I healthy volunteers ($n = 13$) receiving a 9 mg/kg dose

Fig. 3 Model fit for healthy volunteers. Healthy volunteers received 6, 9, 12, or 15 mg/kg of pyronaridine tetraphosphate. The black line represents the model prediction, while the dots indicate observed concentrations in whole blood. The shaded area depicts the two-fold prediction interval

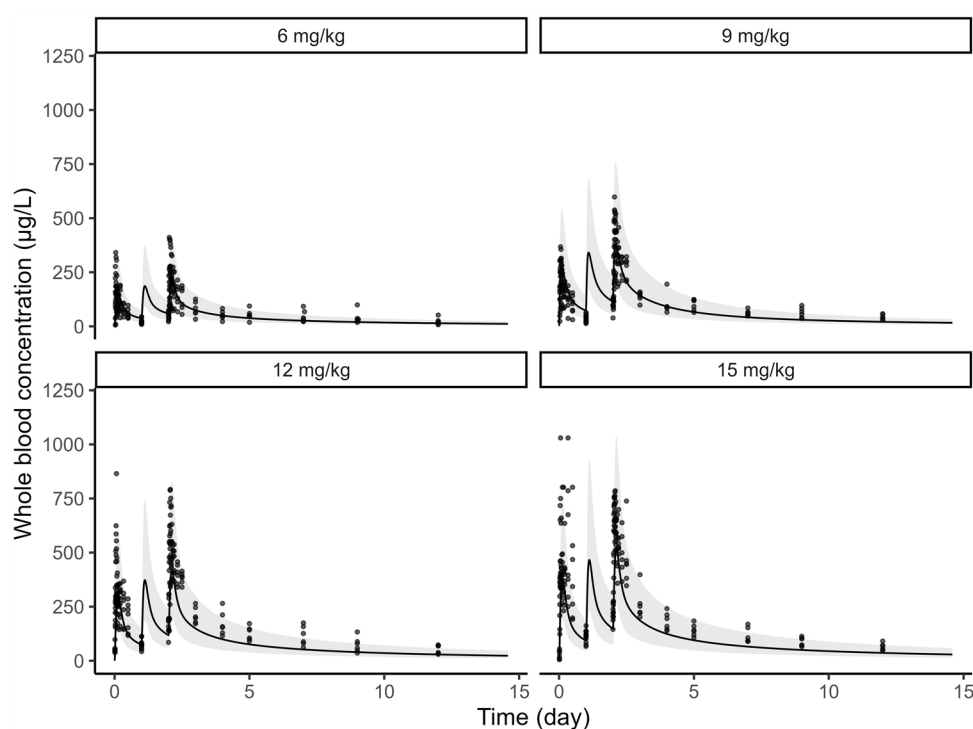
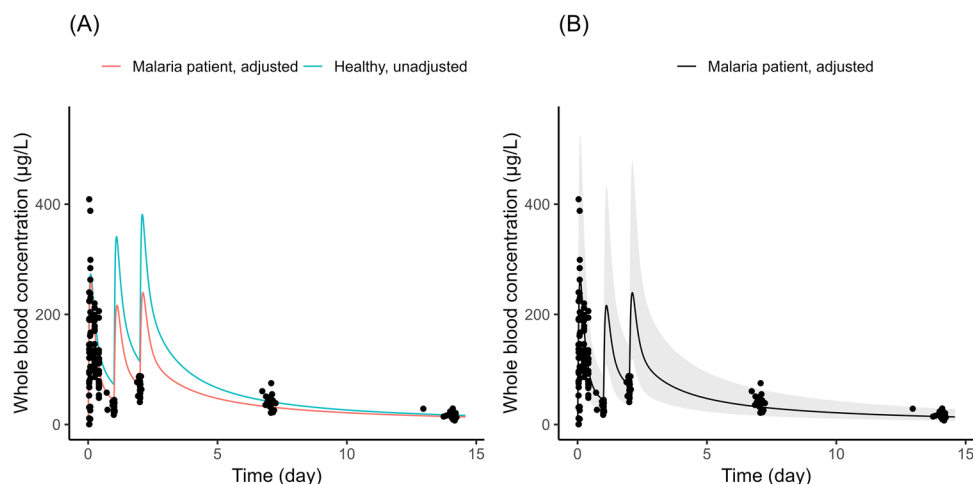


Fig. 4 Effect of malaria infection. **A** Model predictions unadjusted (blue) and adjusted (red) for patient characteristics, including low hemoglobin and drug accumulation in parasites. **B** Final model fit for patients with malaria. Malaria patients received 540 or 720 mg pyronaridine tetraphosphate on the basis of body weight. The black line represents the model prediction, while the dots indicate observed concentrations in whole blood. The shaded area depicts the two-fold prediction interval for the final malaria-adjusted model



0.78, and 0.72, respectively. External validation results are shown in Supplementary Fig. S1.

The model for healthy individuals overestimated the whole-blood concentration-time profile of pyronaridine in malaria patients, especially during the dosing period. Adjusting the individual-specific characteristics to reflect a typical patient with malaria, incorporating decreased Hb (moderate anemia [37]) and an additional parasite compartment improved the model fit (Fig. 4A). Owing to the lack of observations after the second or third dose administration in patients, $K_{p,parasite}$ was fixed at 12.5 to achieve best model fit. The flow rate $Q_{RBC,parasite}$ was fixed at a large value, assuming an instantaneous equilibrium

between parasites and RBCs. In the final model, most observed data fell within the two-fold prediction interval, with an AFE of 1.63 (Fig. 4B). External validation results are shown in Supplementary Fig. S2.

3.3 Target Site Exposure and Effect of Malaria Severity

Simulations of pyronaridine concentrations in whole blood, plasma, RBCs, and parasites are presented in Fig. 5. The total and free drug concentrations in the parasite compartment were more than 10-fold higher

Fig. 5 Predicted pyronaridine concentrations in various compartments. Model predictions of total pyronaridine concentrations (solid line) in whole blood, plasma, red blood cells and parasites, and free drug concentrations (dashed line) in parasites, compared with *in vitro* half maximal inhibitory concentration (IC_{50}) (mean $\pm$ SD [36], shade area). As parasitemia approached zero after treatment, drug concentrations in the parasite compartment were no longer present and therefore not depicted

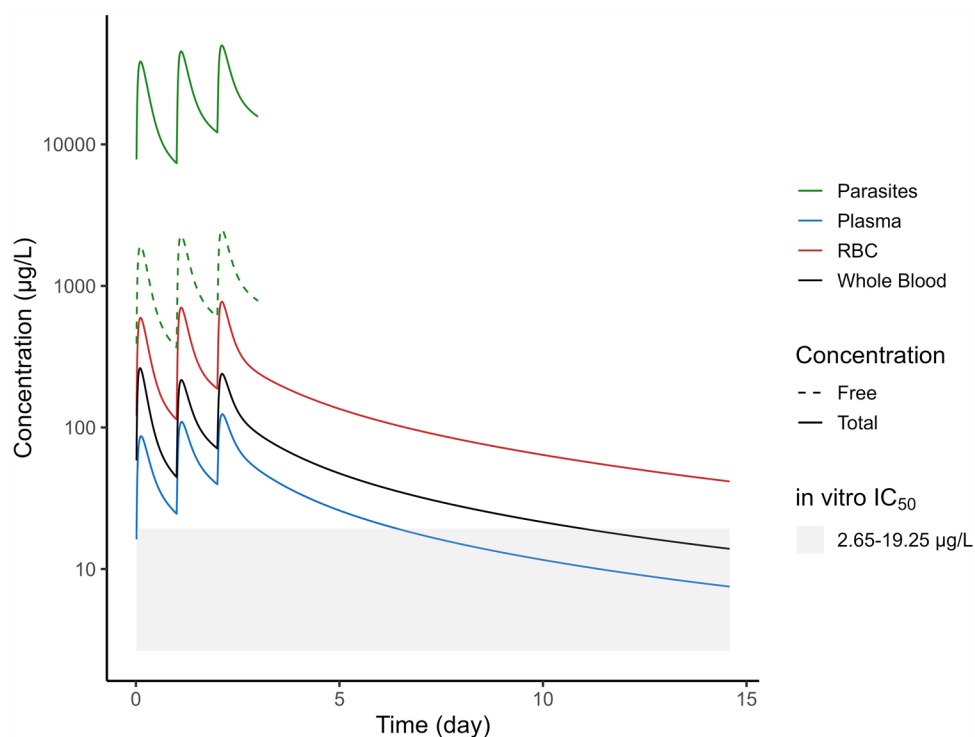
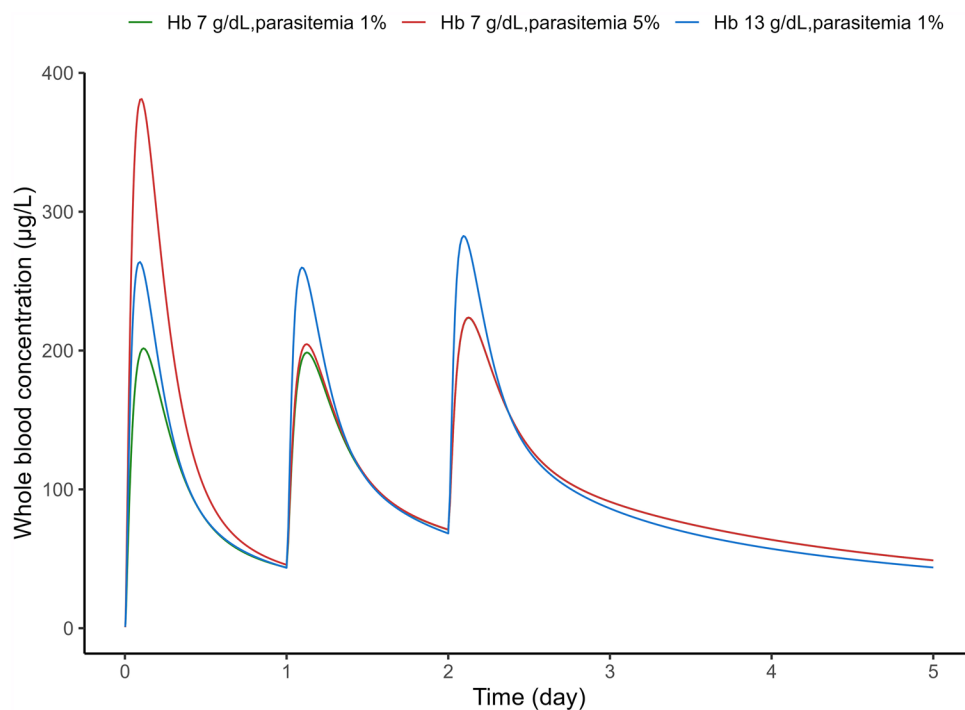


Fig. 6 Simulations illustrating whole blood pyronaridine exposure across varying hemoglobin (Hb) and parasitemia levels



than those in whole blood, indicating extensive target site exposure. Using the plasma concentration as a surrogate for exposure-response assessment may lead to an underestimation of the actual drug effect. This finding supported whole blood as the preferred matrix for evaluating drug exposure and response.

Simulations in patients with varying degrees of malaria-induced Hb reduction and associated parasitemia are provided in Fig. 6 and table 3. Drug exposure following the first dose was strongly influenced by parasitemia level, with C_{max} being 74% higher and $AUC_{0-24h,D1}$ being 53% higher at a parasitemia level of 5% compared with

1%, attributed to drug accumulation within parasites. As parasitemia rapidly declined after treatment, drug exposure following the second and third doses were primarily affected by Hb level, where a 50% reduction in Hb resulting in a 29% decrease in $C_{\max}$ and a 7% decrease in $AUC_{0-24h,D3}$.

4 Discussion

This study employed a PBPK modeling approach to elucidate the critical role of lysosomal sequestration in driving the extensive tissue distribution of pyronaridine. It also provided insights into the substantial differences in drug exposure observed between healthy individuals and patients with malaria during the dosing phase. Furthermore, this study explored the impact of Hb levels and parasitemia on the PK of pyronaridine, identifying both as key determinants of variability in drug exposure throughout the dosing period.

Our findings support lysosomal ion trapping as the primary mechanism driving the extensive distribution of pyronaridine. Lipophilic basic drugs with pKa values that allow partial ionization at intracellular pH can diffuse across lysosomal membranes in their uncharged form. Once inside the lysosome, where the pH ranges around 4.51, these drugs become protonated and are effectively trapped [5]. This mechanism leads to high drug concentrations in lysosome-rich tissues, such as the liver and lungs. Lysosomal trapping has gained attention for its role in repurposing antimalarial drugs, including chloroquine and pyronaridine [6, 42], for treating respiratory infections including coronavirus disease 19 (COVID-19) [43]. Achieving high drug exposure in the lungs is crucial for addressing these conditions, making lysosomal trapping a beneficial factor in attaining and maintaining effective tissue target-site concentrations.

Lysosomal sequestration of pyronaridine provides a plausible explanation for its high apparent volume of distribution, as the drug rapidly accumulates in tissues following administration. This sequestration may enhance efficacy for lysosome-targeted actions but may also increase the risk of localized toxicity. For example, elevated levels of liver transaminases have been observed in previous studies following the use of pyronaridine [44]. Furthermore, the slow release of sequestered pyronaridine from lysosomes into the cytoplasm contributes to its prolonged systemic presence, with detectable drug levels persisting in whole blood for at least 42 days after a 3-day treatment [15]. This extended exposure may underlie its long-acting therapeutic potential in preventing malaria recurrence [3].

Pyronaridine exhibits a strong affinity for RBCs, with the reported BP_{ratio} ranging from 1.6 to 15 in humans. This high and variable BP_{ratio} reflects its preferential partitioning into blood cells. Accordingly, the distribution of pyronaridine

is hypothesized to be influenced by Hb levels, as observed with other high BP_{ratio} drugs such as tacrolimus [45]. Our model supports this hypothesis, identifying changes in Hb as a key factor explaining the observed differences in whole blood exposure between healthy volunteers and patients with malaria. Although plasma protein binding was not explicitly assessed in this study, disease-related alterations in protein levels may also contribute to variability in systemic exposure and warrant further investigation. In addition, pyronaridine concentrations in whole blood may partly reflect accumulation in lysosome-rich cells such as white blood cells. However, in the context of malaria, changes in red blood cell physiology are likely the predominant driver of altered pyronaridine distribution.

Pyronaridine preferentially accumulates within the food vacuole of *Plasmodium* parasites and demonstrates a strong affinity for hemozoin, which is an important disposal product of the heme detoxification process by the parasite [10]. This interaction may enhance drug retention in RBCs, particularly in patients with high parasite burdens, as infected erythrocytes sequester the drug. However, malaria infection also leads to RBC destruction, causing anemia. The resulting reduction in RBCs may limit pyronaridine trapping, potentially lowering its overall retention in whole blood. Our simulations (Fig. 6) supported that the parasitemia level substantially influenced AUC_{0-24h} following the first dose. However, owing to the rapid parasite reduction after drug treatment, changes in Hb have a greater impact on AUC_{0-24h} after the second and third doses. The opposite effects of parasitemia and Hb level, both associated with malaria infection, may explain the high variability observed in the $C_{\max}$ and AUC during the dosing period.

Our model simulations supported the use of whole blood for PK assessments for pyronaridine and highlights the potential risk of underestimating drug effects when relying on plasma measurements. Whole blood has been the preferred matrix for PK studies, as pyronaridine primarily accumulates within RBCs. Consequently, whole blood measurements are believed to provide a more accurate representation of the drug's distribution and elimination. In contrast, plasma assays have been shown to inadequately capture this intracellular distribution, leading to an underestimation of total drug concentrations [46]. Measuring drug concentrations in whole blood enables the use of dried blood spot (DBS) sampling, which offers practical advantages such as simplified sample collection, preparation, and storage, particularly beneficial in resource-limited settings [47].

This study has several limitations. First, the model was developed and validated using a limited dataset, which may constrain the generalizability of the findings. Second, while Hb levels were incorporated to account for malaria-induced

physiological changes, other potentially relevant factors, such as alterations in hepatic enzyme activity, plasma protein binding, or organ perfusion, were not explored, as Hb was considered the most mechanistically plausible driver of the observed changes in pyronaridine distribution. Third, parasitological data during the treatment period and artesunate PK data were not available for model development. This precluded a direct, individual-level evaluation of the interplay between drug exposure and parasite dynamics during the early treatment phase. To address this, we relied on published parasite clearance values for artesunate to inform our analysis.

In conclusion, this study demonstrates the use of PBPK modeling in elucidating key PK mechanisms that are difficult to capture using population PK modeling approach. Specifically, it highlights the pivotal roles of lysosomal sequestration and variations in parasitemia and Hb levels in the distribution of pyronaridine. These factors may strongly impact drug accumulation at the target site, which is crucial for assessing treatment response in malaria.

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Declarations

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Competing Interests J.S. is an employee of Shin Poong Pharm. Co. Ltd, which provided the study drug. Alwin Huitema is an Editorial Board member of Clinical Pharmacokinetics. Alwin Huitema was not involved in the selection of peer reviewers for the manuscript nor any of the subsequent editorial decisions. All other authors declare that they have no competing interests relevant to the content of this article.

Availability of Data and Material The raw data are available upon reasonable request by an email to the corresponding author.

Code Availability The model files are provided as supplementary material. Other codes are available upon reasonable request.

Consent for Publication The authors affirm that human research participants provided informed consent for publication of their data.

Authors' Contributions U.D.A., K.K., T.D., H.M.M., J.K.T., D.Y., F.A.K., V.M., E.S., A.V., and J.S. contributed to protocol development, study conduct, study registration, recruitment, and data collection. W.S. performed the bioanalysis and measured the study samples. W.C., A.H., and T.D. conducted the PBPK modelling. The manuscript was drafted and reviewed by W.C., A.H., and T.D. All authors reviewed, commented on, and approved the final submitted version of the manuscript.

Ethics Approval/Consent to Participate The PYRAPREG trial (PACTR202011812241529) was conducted in accordance with the principles of the Declaration of Helsinki. Ethical approval was obtained from the following committees: the Ethics Committee for Health Research in Burkina Faso, the National Health Ethics Committee in the Democratic Republic of Congo, the Ethics Committee of the Faculty of Medicine and Odontostomatology/Faculty of Pharmacy in Mali, the Gambia Government/MRCG Joint Ethics Committee, and the National Bioethics Committee for Health in Mozambique. Written informed consent was obtained from all participants prior to their inclusion in the study. A four-part phase I clinical trial (Protocol number SP-C-001-03) was conducted at the Clinical Trial Center of Seoul National University, Seoul, South Korea, in accordance with Good Clinical Practice (GCP) guidelines and the Declaration of Helsinki. Ethical approval was granted by the Institutional Review Board of Seoul National University. Written informed consent was obtained from each participant prior to study participation.

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